SAS: PROCS

- Review of DATA step and some more
- Review of PROC step and some more

Reference: Slides by Professor Papachristou.

REVIEW OF DATA STEP

```
data grades;
   infile datalines dsd;
   input
       subject
       gender $
       exam1
       exam2
       ;
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  86
20, M,  82,  85
14, F,  88,   .
;
```

Data manipulation occurs right after `input`...

Total rows: 5 Total columns: 4

	subject	gender	exam1	exam2
1	10	M	80	84
2	7	M	85	89
3	4	F	90	86
4	20	M	82	85
5	14	F	88	.

Keep/drop columns (≈select)

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   drop subject; /* keep/drop columns */
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  86
20, M,  82,  85
14, F,  88,   .
;
run;
```

Total rows: 5 Total columns: 3 Rows 1-5

	gender	exam1	exam2
1	M	80	84
2	M	85	89
3	F	90	86
4	M	82	85
5	F	88	.

Write columns (≈mutate)

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   examavg = (exam1+exam2)/2;
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  86
20, M,  82,  85
14, F,  88,   .
;
run;
```

Total rows: 5 Total columns: 5 Rows 1-5

	subject	gender	exam1	exam2	examavg
1	10	M	80	84	82
2	7	M	85	89	87
3	4	F	90	86	88
4	20	M	82	85	83.5
5	14	F	88	.	.

Write columns (≈mutate)

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   examavg = mean(exam1, exam2);
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  86
20, M,  82,  85
14, F,  88,   .
;
run;
```

Total rows: 5 Total columns: 5 Rows 1-5

	subject	gender	exam1	exam2	examavg
1	10	M	80	84	82
2	7	M	85	89	87
3	4	F	90	86	88
4	20	M	82	85	83.5
5	14	F	88	.	88

Facts when creating new variables:

1. New variables are initialized to missing
2. Missing values are < any value.
3. Reference missing numeric values by .
4. Reference missing character values by ' ' (with or without space).

Aggregation functions like `mean()`, `sum()`, `n()` ignore missing values by default.

More math operations and functions:

- Exponentiation x^y is written `x**y`
- `abs()`, `log()`, `log10()`, `sqrt()`,
- `round( , n)` rounds to nearest multiple of n (includes decimals)

Write columns with if ... else

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   if exam2 >= 90 then grade = 'A';
   else if exam2 >= 80 then grade = 'B';
   else if exam2 >= 70 then grade = 'C';
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,   .
;
run;
```

	subject	gender	exam1	exam2	grade
1	10	M	80	84	B
2	7	M	85	89	B
3	4	F	90	66	
4	20	M	82	75	C
5	14	F	88	.	

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   if exam2 >= 90 then grade = 'A';
   else if exam2 >= 80 then grade = 'B';
   else if exam2 >= 70 then grade = 'C';
   else if exam2 < 60 then grade = 'F';
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,   .
;
run;
```

NA is less than 60 in SAS, so row 5 gets an F grade.

Total rows: 5 Total columns: 5

	subject	gender	exam1	exam2	grade
1	10	M	80	84	B
2	7	M	85	89	B
3	4	F	90	66	
4	20	M	82	75	C
5	14	F	88	.	F

Write columns with if ... else

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   if exam2 >= 90 then grade = 'A';
   else if missing(exam2) then grade = 'Missing';
   else if exam2 >= 80 then grade = 'B';
   else if exam2 >= 70 then grade = 'C';
   else if exam2 < 60 then grade = 'F';
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,   .
;
```

Total rows: 5 Total columns: 5

	subject	gender	exam1	exam2	grade
1	10	M	80	84	B
2	7	M	85	89	B
3	4	F	90	66	
4	20	M	82	75	C
5	14	F	88	.	M

(Fix truncated characters by writing in more spaces...)

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   if exam2 >= 90 then grade = '      A';
   else if missing(exam2) then grade = 'Missing';
   else if exam2 >= 80 then grade = 'B';
   else if exam2 >= 70 then grade = 'C';
   else if exam2 < 60 then grade = 'F';
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,   .
;
```

(Or declare number of characters needed by the grade column)

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
   length grade $7;
   if exam2 >= 90 then grade = 'A';
   else if missing(exam2) then grade = 'Missing';
   else if exam2 >= 80 then grade = ' B';
   else if exam2 >= 70 then grade = 'C';
   else if exam2 < 60 then grade = 'F';
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,
```

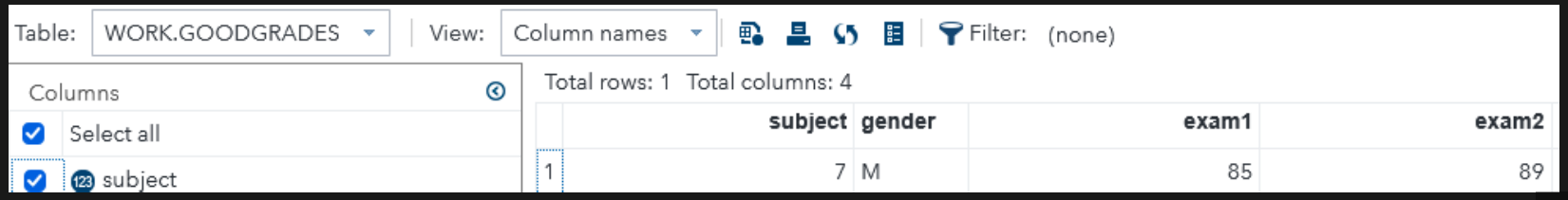
Total rows: 5 Total columns: 5

Rows 1-5

	subject	gender	exam1	exam2	grade
1	10	M	80	84	B
2	7	M	85	89	B
3	4	F	90	66	
4	20	M	82	75	C
5	14	F	88	.	Missing

Filter rows with WHERE

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,   .
;
run;
data goodgrades;
   set grades;
   where exam1 > 80 and exam2 > 80;
run;
```



Filter rows with WHERE

```
data grades;
   infile datalines dsd;
   input subject gender $ exam1 exam2;
datalines;
10, M,  80,  84
 7, M,  85,  89
 4, F,  90,  66
20, M,  82,  75
14, F,  88,   .
;
run;
data goodgrades;
   set grades;
   where exam1 > 80 and exam2 > 80;
run;
```

- `set` helps read in existing SAS dataset for further DATA step manipulation.
- `where` step could not run before data was read in, so do above or replace with if statement.

REVIEW OF PROC STEP

Common PROCs

- PRINT —
- SORT —
- MEANS — Continuous variables
- FREQ — Categorical variables: freq. tables, χ^2-test
- UNIVARIATE — Continuous variables
- GLM — Linear model (later)

Syntax for procedures:

```
PROC PROCNAME DATA=datasetname <options> ;
    substatements/<options> ;
RUN;
```

Common substatements in PROC steps:

- `LABEL` relabels column names
- `FORMAT` changes table entry format
- `VAR` selects variables to analyze
- `WHERE` filters input dataset
- `BY` process data in separate groups, (fails on unsorted groups so use PROC SORT first)
- `CLASS` process data in separate groups (works on fewer PROC than `BY`)

Read in and store SAS dataset `wbc` (white blood cell):

```
filename myurl url "https://manctsui.github.io/datasets/WBC.xl
PROC IMPORT
    DATAFILE=myurl
        OUT = wbc
        DBMS = xlsx
    REPLACE ;
    GETNAMES = yes;
RUN;
```

Total rows: 12 Total columns: 12 Rows 1-12

	Name	State	Bday	Hght	Wght	Sex	SexC	Educ	WBC1	WBC2	WBC3
1	Al	NJ	08/18/57	61	190	0	M	7	12.5	10.3	14.5
2	Jill	NJ	10/07/60	61	135	1	F	2	10.0	13.2	13.8
3	Anna	PA	05/26/62	59	119	1	F	6	13.9	12.5	.
4	Ty	DE	07/06/72	67	142	0	M	.	14.2	12.5	11.4
5	Mary	PA	09/03/74	70	140	1	F	8	10.5	14.5	12.7
6	Ruth	DE	10/07/75	60	110	1	F	3	12.0	13.8	13.2
7	Ada	NJ	11/28/79	63	120	1	F	3	14.6	10.7	14.7
8	Joe	PA	11/10/93	72	180	0	M	1	11.4	13.1	11.8
9	Tim	PA	08/19/99	68	225	0	M	2	11.1	10.4	14.1
10	Kate	DE	06/20/00	65	147	1	F	6	12.7	12.3	13.3
11	Jim	NJ	07/22/01	79	184	0	M	6	14.8	.	14.1
12	Moe	NJ	08/12/02	70	300	0	M	2	14.7	10.1	11.8

Here left-aligned numbers mean the numeric data are read as character strings, so I fixed the xlsx file's "." to empty cell and read it in again.

Obs	Name	State	Bday	Hght	Wght	Sex	SexC	Educ	WBC1	WBC2	WBC3	WBC4
1	Ruth	DE	10/07/75	60	110	1	F	3	12.0	13.8	13.2	11.2
2	Jim	NJ	07/22/01	79	184	0	M	6	14.8	.	14.1	10.9
3	Mary	PA	09/03/74	70	140	1	F	8	10.5	14.5	12.7	14.6
4	Ty	DE	07/06/72	67	142	0	M	.	14.2	12.5	11.4	12.5
5	Moe	NJ	08/12/02	70	300	0	M	2	14.7	10.1	11.8	14.7
6	Ada	NJ	11/28/79	63	120	1	F	3	14.6	10.7	14.7	13.4
7	Al	NJ	08/18/57	61	190	0	M	7	12.5	10.3	14.5	14.9
8	Jill	NJ	10/07/60	61	135	1	F	2	10.0	13.2	13.8	11.2
9	Kate	DE	06/20/00	65	147	1	F	6	12.7	12.3	13.3	15.0
10	Tim	PA	08/19/99	68	225	0	M	2	11.1	10.4	14.1	13.5
11	Joe	PA	11/10/93	72	180	0	M	1	11.4	13.1	11.8	10.3
12	Anna	PA	05/26/62	59	119	1	F	6	13.9	12.5	.	12.7

```
proc print data = wbc(obs=5);
run;
```

Parenthesis lets us do some DATA step options in PROC step. Here obs=5 means we look at first 5 rows or observations:

Obs	Name	State	Bday	Hght	Wght	Sex	SexC	Educ	WBC1	WBC2	WBC3	WBC4
1	Ruth	DE	10/07/75	60	110	1	F	3	12.0	13.8	13.2	11.2
2	Jim	NJ	07/22/01	79	184	0	M	6	14.8	.	14.1	10.9
3	Mary	PA	09/03/74	70	140	1	F	8	10.5	14.5	12.7	14.6
4	Ty	DE	07/06/72	67	142	0	M	.	14.2	12.5	11.4	12.5
5	Moe	NJ	08/12/02	70	300	0	M	2	14.7	10.1	11.8	14.7

```
proc print data = wbc(obs=5);
    where Wght > 150;
run;
```

where filters rows by given condition

Obs	Name	State	Bday	Hght	Wght	Sex	SexC	Educ	WBC1	WBC2	WBC3	WBC4
2	Jim	NJ	07/22/01	79	184	0	M	6	14.8	.	14.1	10.9
5	Moe	NJ	08/12/02	70	300	0	M	2	14.7	10.1	11.8	14.7
7	Al	NJ	08/18/57	61	190	0	M	7	12.5	10.3	14.5	14.9
10	Tim	PA	08/19/99	68	225	0	M	2	11.1	10.4	14.1	13.5
11	Joe	PA	11/10/93	72	180	0	M	1	11.4	13.1	11.8	10.3

```
proc print data = wbc(obs=5);
    where Wght > 150;
    var Name State Wght;
run;
```

where filters rows; var selects columns.

Obs	Name	State	Wght
2	Jim	NJ	184
5	Moe	NJ	300
7	Al	NJ	190
10	Tim	PA	225
11	Joe	PA	180

```
proc print data = wbc(obs=5);
    where Wght > 150;
    var Name State Wght;
    by State;
run;
```

where filters rows; var selects columns; by groupings.

State=NJ

Obs	Name	State	Wght
2	Jim	NJ	184
5	Moe	NJ	300
7	Al	NJ	190

State=PA

Obs	Name	State	Wght
10	Tim	PA	225
11	Joe	PA	180

```
proc print label data = wbc(obs=5);
    title 'WBC dataset';
    label Wght = "Weight"
          Bday = "Birthday";
run;
```

`label` changes printed variable names

WBC dataset

Obs	Name	State	Birthday	Hght	Weight	Sex	SexC	Educ	WBC1	WBC2	WBC3	WBC4
1	Ruth	DE	10/07/75	60	110	1	F	3	12.0	13.8	13.2	11.2
2	Jim	NJ	07/22/01	79	184	0	M	6	14.8	.	14.1	10.9
3	Mary	PA	09/03/74	70	140	1	F	8	10.5	14.5	12.7	14.6
4	Ty	DE	07/06/72	67	142	0	M	.	14.2	12.5	11.4	12.5
5	Moe	NJ	08/12/02	70	300	0	M	2	14.7	10.1	11.8	14.7

PROC FORMAT creates custom formats

```
proc format;
    /* Start categorical variables with $. */
    value $stateF 'PA' = 'Pennsylvania'
                  'DE' = 'Delaware'
                  'NJ' = 'New Jersey';
    /* Good practice to end format names in F. */
    value genderF 0 = 'Male' 1 = 'Female';
run;

PROC print wbc;
    /* Format, when used, end with dot. */
    format state $stateF.;
run;
```

	Name	State	Bday	Hght	Wght	Sex	SexC	Educ	WBC1	WBC2	WBC3	WBC4
1	Ruth	Delaware	10/07/75	60	110	1	F	3	12.0	13.8	13.2	11.2
2	Jim	New Jerse	07/22/01	79	184	0	M	6	14.8	.	14.1	10.9
3	Mary	Pennsylva	09/03/74	70	140	1	F	8	10.5	14.5	12.7	14.6
4	Ty	Delaware	07/06/72	67	142	0	M	.	14.2	12.5	11.4	12.5
5	Moe	New Jerse	08/12/02	70	300	0	M	2	14.7	10.1	11.8	14.7
6	Ada	New Jerse	11/28/79	63	120	1	F	3	14.6	10.7	14.7	13.4
7	Al	New Jerse	08/18/57	61	190	0	M	7	12.5	10.3	14.5	14.9
8	Jill	New Jerse	10/07/60	61	135	1	F	2	10.0	13.2	13.8	11.2
9	Kate	Delaware	06/20/00	65	147	1	F	6	12.7	12.3	13.3	15.0
10	Tim	Pennsylva	08/19/99	68	225	0	M	2	11.1	10.4	14.1	13.5
11	Joe	Pennsylva	11/10/93	72	180	0	M	1	11.4	13.1	11.8	10.3
12	Anna	Pennsylva	05/26/62	59	119	1	F	6	13.9	12.5	.	12.7

PROC FORMAT creates custom formats

```
proc format;
    value $stateF 'PA' = 'Pennsylvania'
                  'DE' = 'Delaware'
                  'NJ' = 'New Jersey';
    value genderF 0 = 'Male' 1 = 'Female';
run;

/* `set` reads in wbc but outputs
   `DATA wbc` writes out to wbc.
   So the format is now applied to the file wbc */
DATA wbc;
    set wbc
    format state $stateF.;
run;
```

PROC MEANS WITH OPTIONS

```
proc means data=wbc;
run;
```

The MEANS Procedure

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
Bday	Bday	12	7967.75	6244.69	-866.0000000	15564.00
Hght	Hght	12	66.2500000	5.9256760	59.0000000	79.0000000
Wght	Wght	12	166.0000000	54.4125654	110.0000000	300.0000000
Sex	Sex	12	0.5000000	0.5222330	0	1.0000000
Educ	Educ	11	4.1818182	2.4420558	1.0000000	8.0000000
WBC1	WBC1	12	12.7000000	1.7220495	10.0000000	14.8000000
WBC2	WBC2	11	12.1272727	1.5264933	10.1000000	14.5000000
WBC3	WBC3	11	13.2181818	1.1548317	11.4000000	14.7000000
WBC4	WBC4	12	12.9083333	1.7026495	10.3000000	15.0000000

```
proc means data=wbc n mean std maxdec=2;
    var wght sex wbc1 wbc2;
run;
```

The MEANS Procedure

Variable	Label	N	Mean	Std Dev
Wght	Wght	12	166.00	54.41
Sex	Sex	12	0.50	0.52
WBC1	WBC1	12	12.70	1.72
WBC2	WBC2	11	12.13	1.53

```
proc means data=wbc n mean std maxdec=2;
    var wght wbc1 wbc2;
    class sexC; /* Group by F vs M */
run;
```

The MEANS Procedure

SexC	N Obs	Variable	Label	N	Mean	Std Dev
F	6	Wght	Wght	6	128.50	14.29
		WBC1	WBC1	6	12.28	1.82
		WBC2	WBC2	6	12.83	1.33
M	6	Wght	Wght	6	203.50	54.17
		WBC1	WBC1	6	13.12	1.67
		WBC2	WBC2	5	11.28	1.41

PROC FREQ WITH OPTIONS

PROC FREQ creates tables aggregating frequencies, here by `sex` and `state` respectively:

```
PROC FREQ DATA=wbc;
    TABLES sex state;
RUN;
```

The FREQ Procedure

Sex

Sex	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	6	50.00	6	50.00
1	6	50.00	12	100.00

State

State	Frequency	Percent	Cumulative Frequency	Cumulative Percent
DE	3	25.00	3	25.00
NJ	5	41.67	8	66.67
PA	4	33.33	12	100.00

PROC FREQ for frequency tables

```
PROC FREQ DATA=wbc;
    /* Options to omit cumulative  freq.& column freq.*/
    TABLES state / nocum nocol;
RUN;
```

The FREQ Procedure

State		
State	Frequency	Percent
DE	3	25.00
NJ	5	41.67
PA	4	33.33

PROC FREQ: 2-way frequency table

```
PROC FREQ DATA=wbc;
TABLES sex*state; /* Use asterisk. */
TITLE 'Cross Tabulation of Sex and State';
RUN;
```

Cross Tabulation of Sex and State

The FREQ Procedure

Frequency
Percent
Row Pct
Col Pct

Table of Sex by State

Sex(Sex)	State(State) DE	NJ	PA	Total
0	1 8.33 16.67 33.33	3 25.00 50.00 60.00	2 16.67 33.33 50.00	6 50.00
1	2 16.67 33.33 66.67	2 16.67 33.33 40.00	2 16.67 33.33 50.00	6 50.00
Total	3 25.00	5 41.67	4 33.33	12 100.00

PROC FREQ: 2-way frequency table in long format

```
PROC FREQ DATA=wbc;
TABLES sex*state / LIST;
TITLE 'Cross Tabulation of Sex and State';
RUN;
```

Cross Tabulation of Sex and State

The FREQ Procedure

Sex	State	Frequency	Percent	Cumulative Frequency	Cumulative Percent
0	DE	1	8.33	1	8.33
0	NJ	3	25.00	4	33.33
0	PA	2	16.67	6	50.00
1	DE	2	16.67	8	66.67
1	NJ	2	16.67	10	83.33
1	PA	2	16.67	12	100.00

PROC FREQ: 2-way frequency table with no row and no col frequencies

```
PROC FREQ DATA=wbc;
TABLES sex*state / NOROW NOCOL;
TITLE 'Cross Tabulation of Sex and State';
RUN;
```

Cross Tabulation of Sex and State

The FREQ Procedure

Frequency Percent	Table of Sex by State			
	State(State)			
Sex(Sex)	DE	NJ	PA	Total
0	1 8.33	3 25.00	2 16.67	6 50.00
1	2 16.67	2 16.67	2 16.67	6 50.00
Total	3 25.00	5 41.67	4 33.33	12 100.00

PROC FREQ: 2-way frequency table, with chi-squared test performed.

```
PROC FREQ DATA=wbc;
TABLES sex*state / CHISQ;
TITLE 'Cross Tabulation of Sex and State';
RUN;
```

Cross Tabulation of Sex and State

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of Sex by State			
	State(State)			
Sex(Sex)	DE	NJ	PA	Total
0	1 8.33 16.67 33.33	3 25.00 50.00 60.00	2 16.67 33.33 50.00	6 50.00
1	2 16.67 33.33 66.67	2 16.67 33.33 40.00	2 16.67 33.33 50.00	6 50.00
Total	3 25.00	5 41.67	4 33.33	12 100.00

Statistics for Table of Sex by State

Statistic	DF	Value	Prob
Chi-Square	2	0.5333	0.7659
Likelihood Ratio Chi-Square	2	0.5412	0.7629
Mantel-Haenszel Chi-Square	1	0.1325	0.7158
Phi Coefficient		0.2108	
Contingency Coefficient		0.2063	
Cramer's V		0.2108	

WARNING: 100% of the cells have expected counts less than 5. Chi-Square may not be a valid test.

Sample Size = 12

PROC FREQ: OTHER USEFUL OPTIONS

- CHISQ performs χ^2-test for 2-way tables.
- `MISSING` includes missing data as separate category
- `LIST` produces data in long-format

Optional options:

- `NOCOL` omits column %
- `NOROW` omits row %
- `NOCUM` omits cumulative frequencies and %
- `NOFREQ` omits frequencies
- `NOPERCENT` omits %